

AEGIS-TWIN

Forensic Incident Report

SEVERITY: CRITICAL

Device Name	Coolant Pump	Device ID	DEV-008
Detector / Zone	8	Timestamp	10 May 2026 03:48:36 UTC
Trust Score	0.0 / 100	Reconstruction MSE	0.62884
Entropy-Shannon Div	0.37707	Attack Pattern	Data Exfiltration — large payloads with elevated

Incident Signature (SHA-256)	040E2460ADE5119CFC3089002D95F9879179BAC48AED85B01ACCFD84853390DF
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1 — Executive Summary

A **CRITICAL** severity anomaly was detected. The device exhibits behaviour strongly inconsistent with its established baseline and must be treated as actively compromised. Immediate isolation and incident response are required.

The Aegis-Twin digital twin engine, using an LSTM autoencoder, observed device **Coolant Pump** (ID: **DEV-008**) in sector **8** at **10 May 2026 03:48:36 UTC**. The trust score fell to **0.0 / 100** and reconstruction error rose to **0.62884**, triggering automated forensic collection and report generation.

2 — Digital Twin Analysis Metrics

The following metrics were recorded by the autoencoder at the time of detection. Elevated MSE and JSD values indicate that the observed traffic distribution diverges significantly from the trained baseline.

Metric	Observed Value	Reference / Threshold
Trust Score	0.0 / 100	Healthy range: ≥ 75
Reconstruction Error (MSE)	0.628837	Threshold: 0.20 (medium) / 0.35 (high) / 0.50 (critical)
Jensen-Shannon Divergence	0.377070	Threshold: 0.30 / 0.50 / 0.70
Derived Severity	CRITICAL	Computed from the above three metrics

3 — Top Anomaly Contributing Features

The features below showed the largest deviation between the observed traffic and the trained baseline. Higher delta values indicate greater anomaly contribution.

Feature	Baseline	Observed	Delta	% Change	Direction
Inter-Arrival Time (IAT)	0.4000	1.0000	0.6000	150.0%	↑ Increased
Payload Entropy	0.4000	1.0000	0.6000	150.0%	↑ Increased
Flow Symmetry	0.6000	0.0000	0.6000	100.0%	↓ Decreased

4 — Digital Evidence Snapshot

Raw feature vectors captured at detection time for archival and chain-of-custody purposes.

Feature	Baseline	Current	Abs. Delta
Packet Size	0.50000	0.00000	0.50000
Inter-Arrival Time (IAT)	0.40000	1.00000	0.60000
Payload Entropy	0.40000	1.00000	0.60000
Flow Symmetry	0.60000	0.00000	0.60000

■ 5 — Identified Attack Pattern

Classified pattern: Data Exfiltration — large payloads with elevated entropy

The classification is derived from heuristic analysis of feature deltas and packet inter-arrival timing distributions. It should be treated as an investigative hypothesis and validated by a human analyst.

■ 6 — Packet Traffic Summary

#	Source	Destination	Protocol	Size (B)	IAT (s)
1	—	—	—	—	0.0104
2	—	—	—	—	0.02
3	—	—	—	—	0.0217
4	—	—	—	—	0.0459
5	—	—	—	—	0.0093
6	—	—	—	—	0.0151
7	—	—	—	—	0.0319
8	—	—	—	—	0.0468
9	—	—	—	—	0.0112
10	—	—	—	—	0.0064
11	—	—	—	—	0.0035
12	—	—	—	—	0.0334

■ 7 — Behavioural Timeline

No threat log entries were recorded at the time of report generation.

■ 8 — Risk Assessment

This incident is rated **CRITICAL**. The combination of very low trust score, high reconstruction error, and significant distribution divergence indicates a strong likelihood of active exploitation or data exfiltration. The device should be treated as **fully compromised** until cleared by forensic investigation.

■ 9 — Recommended Remediation Actions

Priority	Remediation Action
Immediate	Isolate the device from the network (quarantine VLAN or physical disconnect).

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Immediate	Preserve volatile memory: capture a RAM image before any reboot.
Short-term	Collect full packet captures and system/application logs for post-incident analysis.
Short-term	Perform a firmware integrity check and compare against known-good hash.
Short-term	Rotate all credentials, API keys, and revoke active sessions associated with the device.
Long-term	Update IDS/IPS signatures to detect similar behavioural patterns.
Long-term	Review and tighten network segmentation and access-control lists for this sector.
Long-term	Schedule a post-incident review and update the Aegis-Twin baseline after remediation.

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